

Is the Prisoner's Dilemma a Newcomb Problem?

This essay explains Lewis's reduction of the Prisoner's Dilemma to Newcomb's Problem and evaluates the strength of Sobel's response from epistemic asymmetry ultimately concluding that the asymmetry is significant.

According to Newcomb's Problem, an agent must choose between two boxes. He may pick either the box on the left or both of the boxes. To choose only the box on the left is to one-box and to choose both boxes is to two-box. The value of box on the right is set at a fixed value, say, \$1000. The value of the box on the left, however, is dependent on the prediction of another individual, the predictor, who sets the value of the left box depending on their prediction of whether the agent will one-box or not. Let us say that the predictor will set the value of the left box at \$1M if she thinks the agent will one-box, and \$0 if she thinks the agent will two-box.

According to Causal Decision Theory, for which it is assumed that the prediction of the predictor is made before the agent makes their choice, meaning causal independence between the prediction and the choice, two-boxing strictly dominates over one-boxing. This is because of the extra \$1000 dollars which is present in either case of the predictor's prediction.

According to Evidential Decision Theory, however, it is assumed that the choice made by the agent is strong evidence of the predictor's prediction. Since this theory produces expected payoffs, one cannot draw the conclusion that either of the choices dominate over the other. Instead, one-boxing yields a much greater expected payoff than two-boxing – therefore one-boxing is preferred.

According to prisoner's dilemma, two prisoners, who cannot communicate with each other, must choose to either cooperate or defect. If both prisoners cooperate, they both get 1 year in prison. If one of the prisoners defects while the other cooperates, the prisoner which cooperated gets 10 years while the one that defected does not get any years. And in the case that both defect, they both get 6 years. Since each prisoner is self-interested and wishes to minimise their own sentence, it is rational for them to only ever choose to defect. This is because, for each prisoner, their sentence for either case presented by their fellow prisoner, their cooperation or defection, is minimised by his own defecting. Therefore, defection dominates cooperation. Since this game involves two players, unlike Newcomb's Problem, we identify defect-defect as a Nash equilibrium – meaning that no player can benefit by unilaterally changing their own strategy. Despite the preferred outcome being achieved only in the case that both prisoners cooperate, individual incentives place the Nash equilibrium at a less preferable position.

Lewis's Inessentials

Lewis argues that, once inessential aspects of Newcomb's Problem are ignored, such as prediction-making process, its structure resembles that of the Prisoners' Dilemma (Lewis, 1979). Lewis modifies the sentence outcomes of the Prisoners' Dilemma such that they resemble the monetary outcomes of Newcomb's Problem. If both prisoners defect, they get \$1000. If they differ in their choices, the defector gets \$1M + \$1000 and the cooperator does not receive anything. And if both prisoners cooperate, they get \$1M. Lewis then asserts that the decision problem for both games is defined by the following three points:

1. Players in the games are offered \$1000,
2. A player's \$1M outcome is casually independent of what he chooses – Lewis accepts Causal Decision Theory,
3. That \$1M outcome can only be acquired by that same player if and only if the other does not take their \$1000 – if they do not defect,

with the exception that, for Newcomb's Problem, the \$1000 is based on a prediction that the player does not take their small reward rather than another player acting in their self-interest as is the case for the Prisoners' Dilemma. Lewis calculates that, even according to Evidential Decision Theory, the prediction must only have an accuracy of 0.5005 to favour one-boxing. Assuming that the average person is likely reliable enough for Newcomb's Problem, reliability is inessential for the reducibility of the Prisoners' Dilemma to Newcomb's Problem. Since it is also inessential how that prediction is produced, we can also consider the choice of another player, in another Newcomb's game, to also predict the choices of other players in other games. Therefore, Lewis concludes for both games:

4. The large reward is received if and only if an approximately 50% reliable predictive process predicts that the player does not take their small reward.

Sobel's Epistemic Asymmetry

Sobel objects to the reducibility of Prisoners' Dilemmas to Newcomb's Problems for cases in which the prisoners have a high degree of certainty (Badici, 2023). Unlike Lewis, who considers the prediction-making process to be inessential, Sobel argues these 'near-certainty' cases, present an epistemic asymmetry which makes the reduction invalid. According to Sobel, near-certain cases of Newcomb's Problem are identified by the player's choice. Whereas for the Prisoners' Dilemma, the near-certainty may come from the two players being *psychological twins* or from having come to an agreement prior to playing the game.

Even if this distinction was not inessential, I see no reason why the same cannot be said for Newcomb's Problem. If the predictor happened to recognise his good friend, and knowing him, would be near-certain of his choice, would not his 'prediction' benefit in the same way that the prisoners would if they came to some agreement before playing the game? I see no reason why the predictor, being able to form his prediction with some privileged information, is any more in violation of the rules than two prisoners who cheat by communicating with each other. I also see no reason to exclude the possibility of psychological twins from Newcomb's Problem if we can entertain it for the Prisoners' Dilemma. If these cases are possible for both games, then there is no epistemic asymmetry between them.

Anti-reductionists may respond that the structural difference lies in epistemic asymmetry lies in the prisoners' choices being casually independent by default, so long as external influences are not present. However, the predictor's prediction must positively correlate with the choice, his choice of one-box or two-box is not casually-independent with the agent's by default. Therefore, reducible Prisoners' Dilemmas must be cases in which the external factors, such as being psychologically twins or having come to some agreement prior to playing the game, are the same ways in which the predictor for Newcomb's Problem produces his own predictions.

Therefore, the structure of a Prisoners' Dilemma does permit reduction to Newcomb's Problem, but it must be modified to account for the positive correlation of the predictor's predictions. In general, a Prisoners' Dilemma is not reducible to Newcomb's Problem.

References

Badici, E. (2023). Prisoner's Dilemma and Newcomb's Problem: Two Problems or One? *Philosophia*.

Lewis, D. (1979). Prisoners' Dilemma is a Newcomb Problem. *Philosophy & Public Affairs*.